PRIMARY 380 VDC (HV_* nets: 2.5 mm clearance, isolated from secondary)

SECONDARY 50 V (pulse current flows bank <–> backplane through the studs, not this PCB)

HV_IN_P
HV_N
J1
380VDC IN
HV_IN_P
6.3A 500V
HV_F
HV_F
10R NTC
RT1
HV_P
HV_P
C1
10u 450V
HV_N
HV_P
C2
10u 450V
HV_N
+12V
HV_P
A1
14
K1
G2RL-1A 12V
A2
13
RLY_DRV
HV_F
ON1
470R1
ON_A1
U3 PCB17
HV_ONOFF1
GND
2
3
HV_N
HV_P
+IN
ON/OFF
-IN
U1
+OUT
+S
TRIM
SHARE
-S
-OUT
380V >50V 600W
BRICK1_OUT
VOUT_50V
TRIM1
SHARE
GND
ON2
470R2
ON_A2
U4 PCB17
HV_ONOFF2
GND
2
3
HV_N
HV_P
+IN
ON/OFF
-IN
U2
+OUT
+S
TRIM
SHARE
-S
-OUT
380V >50V 600W
BRICK2_OUT
VOUT_50V
TRIM2
SHARE
GND
VOR
VOUT_50V
R7
0R
C3
1000u 63V
C4
1000u 63V
C5
1000u 63V
C6
1000u 63V
C7
10u 100V
GND
VOUT_50V
J2
+50V STUD
GND
J3
0V STUD
+3V3
VOR
VOUT_50V
U9
INA240A1
VIN+
VIN-
VS
REF
OUT
GND
5
I_OUT
R8
1R
I_OUT
C8
10n
GND
VOUT_50V
R9
200R
VOUT_MON
VOUT_MON
R10
47K
GND
TRIM1
R11
47K
GND
VOR
D1
100V 60A
BRICK1_OUT
TH1
10k NTC
GND
NTC1
R5
100R
NTC1
+3V3
TRIM2
R12
47K
GND
VOR
D2
100V 60A
BRICK2_OUT
TH2
10k NTC
GND
NTC2
R6
100R
NTC2
+3V3
RS485_A
RS485_B
J5
RS485
GND
+3V3
U8
THVD1450
RS485_RO
RS485_DE
RS485_DE
RS485_DI
RO
RE
DE
DI
VCC
GND
A
B
5
RS485_A
RS485_B
+3V3
U7
STM32G031K8T6
VDD
VSS
NRST
PF2
PC6
PC14
PC15
RLY_EN
LED_DRV
RS485_DE
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9/PA11
PA10/PA12
PA13
PA14
PA15
VOUT_MON
I_OUT
RS485_DI
RS485_RO
NTC1
NTC2
ON1
ON2
SWDIO
SWCLK
GND
+3V3
J6
SWD
NRST
C14
100n
LED_DRV
R13
47R
LED_A
D4
RN
GND
PWR_FLAG
GND
PWR_FLAG
+12V
PWR_FLAG
HV_N
PWR_FLAG
HV_P
PWR_FLAG
VOUT_50V
RLY_DRV
D3
SM4148W
+12V
RLY_DRV
Q1
2N7002K
1
2
3
RLY_G
R14
100R
RLY_G
GND
VOUT_50V
U5
+IN
ON/OFF
-IN
+OUT
TRIM
-OUT
+12V
GND
GND
48V >12V 50W
+12V
C9
22u 25V
GND
J4
12V OUT
+12V
U6
LM1117-3.3
VI
VO
GND
3
2
1
+3V3
GND
+12V
C10
10u
GND
+3V3
C11
10u
GND
+3V3
C12
100n
GND
+3V3
C13
100n
GND
H1
M4
GND
H2
M4
GND
H3
M4
GND
H4
M4
GND

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SENTINEL-UHF reference design

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File: SENTINEL_PSU.kicad_sch

Title: SENTINEL-UHF Subarray Power Supply, 380 V -> 50 V

Size: A2	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 1/1